

DETAILED USE CASES

Elaboration Iteration 2

Use case name:	Login to System	
Scenario:	A user authenticates to gain access to the POS system.	
Triggering event:	User needs to access the system.	
Brief description:	The user enters a username and password; the system validates the credentials, authenticates the user, and grants access according to their assigned role.	
Actors:	Cashier, Administrator	
Related use cases:	All other use cases (precondition for most)	
Stakeholders:	Cashier, Administrator, Owner	
Preconditions:	User has a valid, active account.	
Postconditions:	User is authenticated and granted access appropriate to their role.	
Flow of activities	Actor	System
	1) Enters username and password 2) Submits	1.1) Validate credentials 1.2) Authenticate and determine role 1.3) Grant role-based access
Exception conditions:	Invalid credentials; disabled account.	

Use case name:	Add Product	
Scenario:	Authorized staff add a new electronic product/gadget to inventory.	
Triggering event:	A new product needs to be listed for sale.	
Brief description:	The administrator enters product details and an initial stock quantity; the system validates the input and creates the Product record together with its initial Inventory record.	
Actors:	Administrator / Inventory Staff	
Related use cases:	Update Product Information	
Stakeholders:	Administrator, Owner	
Preconditions:	Administrator is logged in.	
Postconditions:	Product (and its Inventory record) is created.	
Flow of activities	Actor	System
	1) Selects Add Product 2) Enters product details (name, brand, price, barcode, description, initial stock) 3) Submits	1.1) Display product entry form 1.2) Validate input 1.3) Create Product record and initial Inventory record
Exception conditions:	Duplicate barcode; invalid/missing required fields.	

Use case name:	Update Product Information	
Scenario:	Authorized staff update product details.	
Triggering event:	Price, description, or stock details need to change.	
Brief description:	The administrator selects an existing product, edits its details, and saves the changes; the system validates the updates and writes them to the product record.	
Actors:	Administrator / Inventory Staff	
Related use cases:	Add Product	
Stakeholders:	Administrator, Owner	
Preconditions:	Product exists.	
Postconditions:	Product record is updated.	
Flow of activities	Actor	System
	1) Selects product 2) Edits details 3) Saves	1.1) Retrieve current product details 1.2) Validate changes 1.3) Update product record
Exception conditions:	Invalid data entered.	

Use case name:	Remove Product	
Scenario:	The administrator removes a discontinued product.	
Triggering event:	A product is no longer sold by the store.	
Brief description:	The administrator selects a product and confirms removal; the system marks the product as discontinued or removes it so that it is no longer sellable.	
Actors:	Administrator	
Related use cases:	Update Product Information	
Stakeholders:	Administrator, Owner	
Preconditions:	Product exists.	
Postconditions:	Product is removed (or marked discontinued) and no longer sellable.	
Flow of activities	Actor	System
	1) Selects product 2) Confirms removal	1.1) Prompt for removal confirmation 1.2) Mark/remove product record
Exception conditions:	Product has an active/pending sale referencing it — system should flag rather than hard-delete.	

Use case name:	Manage User Accounts	
Scenario:	The administrator creates, updates, or disables user accounts.	
Triggering event:	Staffing changes (new hire, role change, departure).	
Brief description:	The administrator opens user management, then creates, edits, or disables an account and assigns the appropriate role; the system validates the input and updates the account record.	
Actors:	Administrator	
Related use cases:	Login to System	
Stakeholders:	Administrator, Owner	
Preconditions:	Administrator is logged in.	
Postconditions:	User account is created, updated, or disabled with the correct role.	
Flow of activities	Actor	System
	1) Selects Manage Users 2) Creates/edits/disables an account, assigns a role	1.1) Display user list 1.2) Validate input 1.3) Create/update account record
Exception conditions:	Username already taken; cannot disable the last remaining administrator account.	

Use case name:	View Sales Report	
Scenario:	The administrator views sales performance over a period.	
Triggering event:	Administrator needs sales insight (daily/weekly/monthly).	
Brief description:	The administrator selects a reporting period; the system queries sales records for that period, aggregates totals, and displays the sales report.	
Actors:	Administrator	
Related use cases:	Export Sales Data	
Stakeholders:	Administrator, Owner	
Preconditions:	Sales data exists for the requested period.	
Postconditions:	Sales report is displayed.	
Flow of activities	Actor	System
	1) Selects report period (day/week/month) 2) Requests report	1.1) Query sales records for period 1.2) Aggregate totals, generate report 1.3) Display report
Exception conditions:	No sales data for the selected period.	

Use case name:	View Inventory Report	
Scenario:	Staff view current stock levels across all products.	
Triggering event:	Need to check stock status.	
Brief description:	The administrator requests an inventory report; the system queries current stock levels for all products and displays the report.	
Actors:	Administrator / Inventory Staff	
Related use cases:	Receive Low Stock Alert	
Stakeholders:	Administrator, Owner	
Preconditions:	None.	
Postconditions:	Inventory report is displayed.	
Flow of activities	Actor	System
	1) Requests inventory report	1.1) Query current stock levels for all products 1.2) Display report
Exception conditions:	None.	

Use case name:	Receive Low Stock Alert	
Scenario:	The system automatically alerts staff when stock falls below a threshold.	
Triggering event:	A sale reduces a product's stock below its reorder level.	
Brief description:	After each completed sale the system checks the product stock level against its configured reorder level and generates an alert to staff if the stock has fallen below the threshold.	
Actors:	System (initiator); Administrator / Inventory Staff (recipient)	
Related use cases:	View Inventory Report, Process Sale	
Stakeholders:	Administrator, Owner	
Preconditions:	Reorder level is configured for the product.	
Postconditions:	Staff are notified of the low-stock condition.	
Flow of activities	Actor	System
	(System-initiated, following a sale)	1.1) Check stock level after each sale 1.2) Compare against reorder level 1.3) Generate and send alert if below threshold
Exception conditions:	Reorder level not configured for the product — no alert can be generated.	

Use case name:	Export Sales Data	
Scenario:	The administrator exports sales data for external analysis or record keeping.	
Triggering event:	Need for offline/external analysis, backup, or audit.	
Brief description:	The administrator selects Export together with a period and file format; the system compiles the sales data, generates the export file, and provides it for download.	
Actors:	Administrator	
Related use cases:	View Sales Report	
Stakeholders:	Administrator, Owner	
Preconditions:	Sales data exists for the requested period.	
Postconditions:	An export file is produced.	
Flow of activities	Actor	System
	1) Selects Export and a period/format 2) Confirms	1.1) Compile sales data for the period 1.2) Generate export file 1.3) Provide file for download
Exception conditions:	Export generation fails (e.g. storage/service error).	